

Table 1: Kuo Quadrupole R0500 (fitting region: $\pm 20 \text{ } \mu\text{m}$, 21×21 grid, $z = 0$ plane)

$$\text{Error} = \frac{\text{RMSE}}{|\mathbf{B}_{FEM}|} \cdot 100$$

Kuo Quadrupole (Error=1.35%)		
Pole	$ \ell \text{ } [\mu\text{m}]$	Tip→Charge $[\mu\text{m}]$
P1	468.9	−31.1
P2	469.2	−30.8
P3	469.2	−30.8
P4	468.9	−31.1
Average	469.0	−31.0
R_a	$1.57 \times 10^9 \text{ A/Wb}$	

Tip→Origin = $500 \text{ } \mu\text{m}$

Tip→Charge = $|\ell| - 500$

$N_c = 4$ (4 turns per pole), 4-fold symmetric quadrupole, single-plane fit.